

ARIMA Model Fitting and Analysis of S&P 500 (2010–2025)

1. Data Retrieval and Preprocessing

The Adjusted Daily closing prices for the S&P 500 index between 2010–2025 were retrieved from `quantmod` library, which had a total of **3983** values using:

```
getSymbols("^GSPC",src="yahoo",from = "2010-01-01",to= "2025-11-01")
Sp500_prices<- Ad(GSPC) #adjusted closing price for cleaner, adjusted
series
```

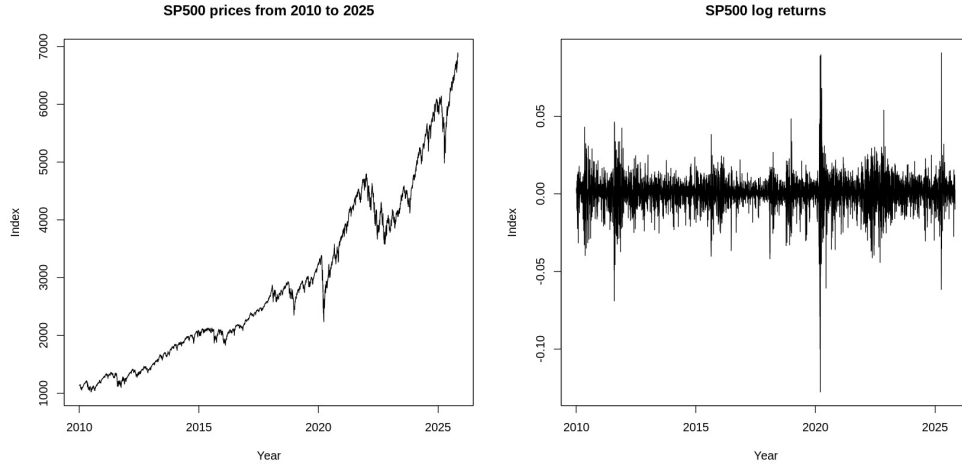


Figure 1: S&P 500 Daily price Vs Log returns

Since raw prices exhibit unstable mean and variances, log differences help stabilize variance, remove trends, resulting in a series that fluctuates around a constant mean crucial for ARIMA model fitting and forecasting as shown in above figure. Daily log-returns are calculated as:

$r_t = \ln(P_t) - \ln(P_{t-1})$ and possible NA values are omitted:

```
Sp500_returns<-diff(log(Sp500_prices)) #log-difference transformation
Sp500_returns<-na.omit(Sp500_returns) #dropping NA values
```

2. Preliminary Analysis and Model Identification

Stationary data is a pre-requisite for an ARIMA model which is tested using Augmented Dickey Fuller (ADF) test. It considers:

H_0 : The series is non-stationary, H_A : The series is stationary

If the **p-value** < **0.05**, null hypothesis is rejected and the series is said to be stationary. ADF tests are conducted using `adf.test()` method of `tseries` library.

```
adf.test(Sp500_prices) #stationarity check on prices
adf.test(Sp500_returns) #stationarity check on log returns
```

Statistic	Sp500_prices	Sp500_returns
p-value	0.9537 > 0.05 (non-stationary)	0.01 < 0.05 (stationary)

Table 1: ADF Test p-values for Price and Return Series

The test confirmed that prices initially had some time-dependent structure and inconsistent variance over time, but log-differencing to obtain log-returns made it stationary.

Autoregressive(AR) Model is a model where the current value(X_t) is based on the previous value(X_{t-1}) plus a random error(ϵ_t). Mathematically, for previous p values:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \epsilon_t$$

Moving Average(MA) Models is a model where the forecast value(y_t) depends on the past forecast errors(ϵ_t). Mathematically, for last q errors:

$$y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q}$$

To determine possible AR(p) and MA(q) component orders, ACF/PACF plots were examined made using:

```
acf(Sp500_ret, main='ACF Plot', col='blue')
pacf(Sp500_ret, main='PACF Plot', col='red')
```

Both plots seem to gradually tail-off, suggesting presence of both AR and MA components. Hence, we assume a simple ARIMA model of order (1,0,1) on the stationary data.

For automated selection, we use `auto.arima()` from `forecast` package, which automatically determines the order (p,d,q) by minimizing information criterion like Akaike Information Criterion(AIC) or Bayesian Information Criterion(BIC).

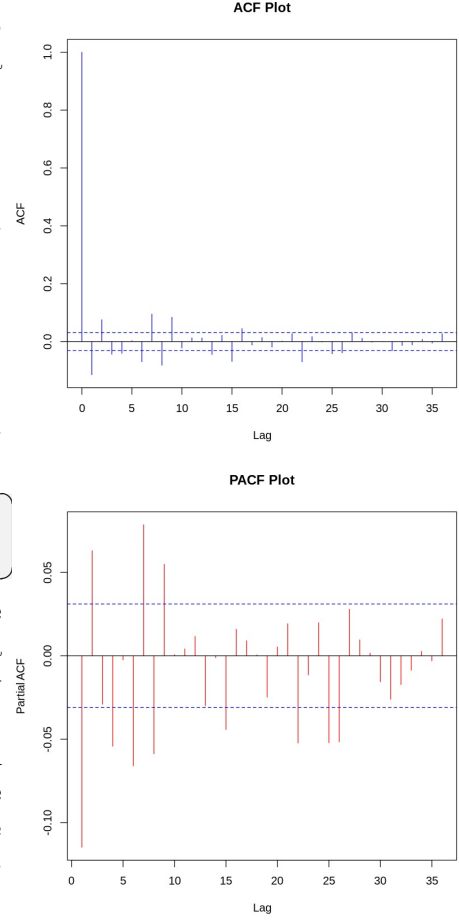


Figure 2: ACF and PACF of Differenced Series

3. ARIMA Model Fitting and Comparison

ARIMA model integrates Autoregression(AR), differencing for stationarity(I), and Moving Average(MA) terms. A general ARIMA(p,d,q) is represented as:

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \theta_1 \epsilon_{t-1} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$$

Before fitting the ARIMA models, data is divided into two contiguous parts separated by a cut-off date- Dec 30, 2022 to imitate a real forecasting scenario: In-sample for model-estimation, parameter selection, diagnostics, and Out-sample for evaluating forecast performance on unseen data.

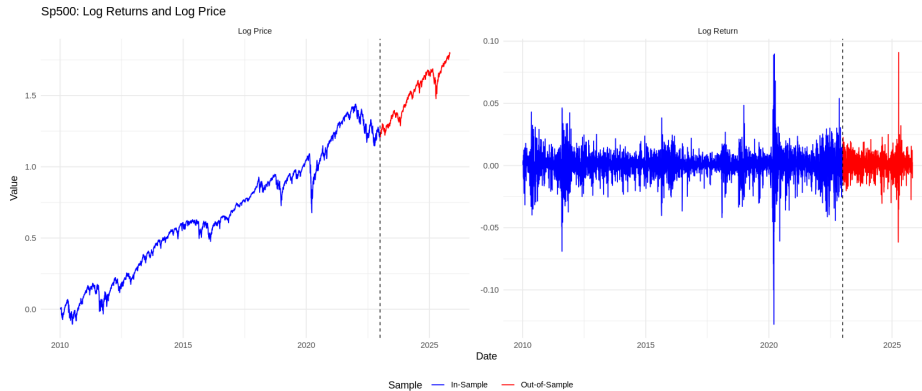


Figure 3: Sp500: Log Returns and Log Price (In-sample and Out-sample)

ARIMA models are fitted manually as well as automatically on In-sample data to obtain model metrics using:

```
cutoff_date<-as.Date("2022-12-30") #cutoff date
in_sample<-Sp500_ret[index(Sp500_ret)<=cutoff_date] #within cutoff date
out_sample <- Sp500_ret[index(Sp500_ret)>cutoff_date] #outside cutoff date
manual_fit<-Arima(in_sample, order = c(1,0,1)) #fitting manual model
auto_fit<-auto.arima(in_sample, d=0) #fitting automatic model
```

Since the data is already stationary, $d = 0$ is set to prevent over-differencing for `auto.arima()`. Final order given was (1,0,0). Comparing both model metrics as:

Model	$\hat{\sigma}^2$	Log-likelihood	AIC	BIC	RMSE	MAE	MASE	Residual ACF1
Manual(1,0,1)	0.0001239	10,064.33	-20,120.66	-20,094.8	0.01112	0.00733	0.6667	0.0094
auto.arima(1,0,0)	0.0001244	10,059.06	-20,112.12	-20,092.7	0.01115	0.00735	0.6682	0.0123

Table 2: Comparison of manual ARIMA(1,0,1) vs `auto.arima()`.

Overall, ARIMA(1,0,1) has comparatively lower AIC/BIC values and error metrics, suggesting better overall fit and predictive accuracy to the in-sample data. Residual ACF near-zero value suggests residuals are close to white noise, indicating a well-fitted model.

4. Residual Diagnostics and Assumption Validation

Residual diagnostics are done to verify that residuals behave as white noise, using Ljung-box test which considers:

H_0 : Series has no serial autocorrelation, H_A : Series has serial correlation

It is conducted using the following code:

```
checkresiduals(manual_fit)
checkresiduals(auto_fit)
```

Models	Ljung-Box Q*	Ljung-Box p-values	Interpretation
manual(1,0,1)	578.92(df = 8)	$2.2e-16 < 0.05$	Residual Autocorrelation
auto arima(1,0,0)	603.24(df = 9)	$2.2e-16 < 0.05$	Residual Autocorrelation

Table 3: Ljung-Box test values for both models

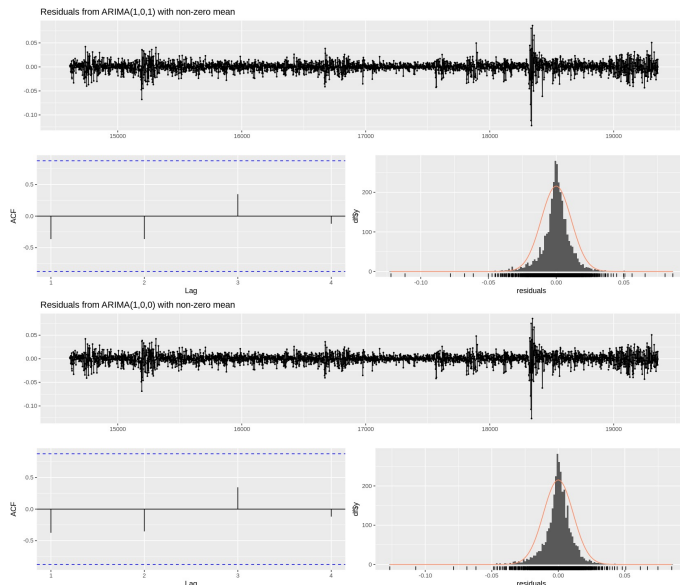
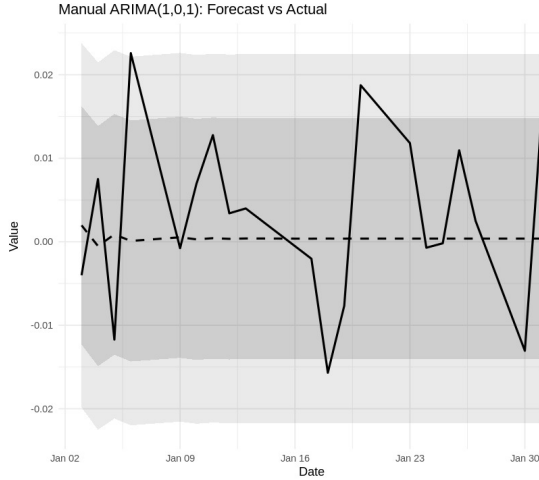


Figure 4: Residual ACF, PACF and Histogram plots

Findings: For both models, there's no evident patterns in residual plots, no significant spikes in ACF plots and histogram plots of residuals look roughly normal but both fail the Ljung-box test indicating residual autocorrelation, which is common for financial returns due to volatility that cannot be captured by ARIMA models, requiring GARCH type models.

5. Model Evaluation and Performance Comparison

For an ARIMA(p,d,q) fitted on the in-sample data with residuals($\hat{e}_t$), the h-step forward forecasts are calculated as: $\hat{y}_{t+h|t} = \mathbb{E}(y_{t+h} | \mathbb{F}_t)$



Forecasting on first 20 Out-sample points, to evaluate performance on unseen data:

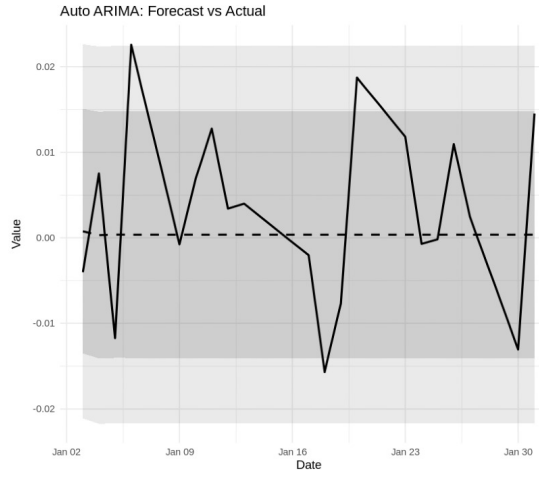
```
manual_forecast<-forecast(manual_fit, h=20)
auto_forecast<-forecast(auto_fit, h=20)
```

Evaluation metrics:

$$\text{RMSE}(h) = \sqrt{\frac{1}{N_h} \sum e_{t+h|t}^2}$$

$$\text{MAE}(h) = \frac{1}{N_h} \sum |e_{t+h|t}|$$

$$\text{MAPE} = \frac{1}{N} \sum \left| \frac{\text{Actual value}_i - \text{Forecasted value}_i}{\text{Actual value}_i} \right| * 100$$



Detailed model metrics on In-Sample(Training) and Out-Sample(Testing) data can be obtained using:

```
manual_accuracy<-accuracy(manual_forecast, out_sample[1:20])
auto_accuracy<-accuracy(auto_forecast, out_sample[1:20])
```

The table below shows the in-sample and out-of-sample forecast performance of both models:

Metric	Manual (1,0,1)	auto.arima (1,0,0)
Training RMSE	0.01112	0.01115
Training MAE	0.007337	0.007354
Training MASE	0.6667	0.6682
Test RMSE	0.01067	0.01055
Test MAE	0.0087	0.0085
Test MASE	0.7918	0.7781
Test MAPE	118.41	114.87

Figure 5: Residual Diagnostics

Diagnostics are done with Ljung-Box test again even for Out-Sample residuals (actual - forecast) because it reveals whether model's predictive assumptions hold in unseen data as well.

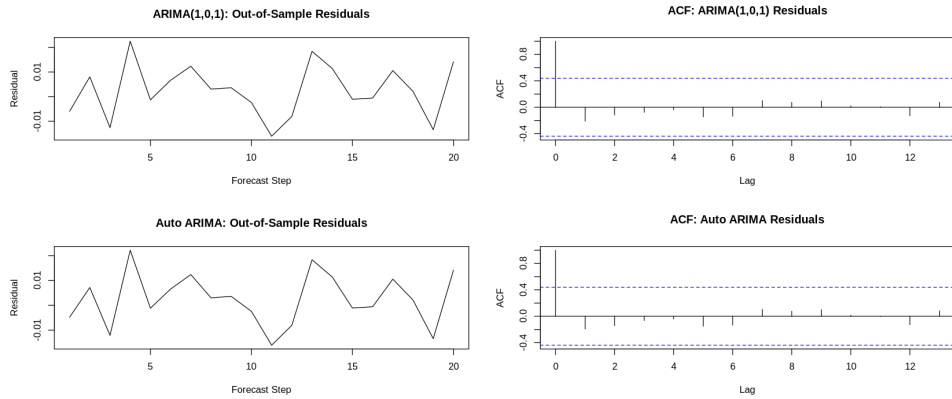


Figure 6: Out-Sample residual diagnostics

Ljung-box test and forecast plots reveal that both manual and auto-arima models had non-significant residuals($p=0.96>0.05$), indicating white noise. Hence, the simpler **auto ARIMA** model with near-identical metrics but marginally better performer on test data is the suitable choice.